PROJECT DATASET BRIEF

Car Price Prediction

Machine Learning Regression Project

1,000 Rows of car data	Columns (7 features + 1 target)	**~£6.7K – £41.8K** Price range (target)	**2000 – 2021** Year range

About the Dataset

This dataset contains detailed information about 1,000 used cars. Each row represents a single vehicle listed for sale. The dataset was sourced from Kaggle and is designed for regression tasks — the goal is to predict the Price of a car based on its other characteristics.

You will use this dataset to build a complete Machine Learning pipeline that takes in a car's features (brand, age, mileage, engine size, fuel type, and transmission) and outputs a predicted selling price.

Dataset Columns Explained

The dataset has 8 columns. The first 7 are input features you will use to build your model. The last column — Price — is your target variable (the value you must predict).

Column	Data Type	Description	Sample Values
`Make`	Categorical	The manufacturer / brand of the car (e.g. Toyota, BMW, Ford, Audi, Honda)	Ford, Audi, BMW, Honda, Toyota
`Model`	Categorical	The variant or trim level labelled as Model A, B, C, D, or E — representing different equipment or configuration levels	Model A, Model B, Model C…
`Year`	Numerical	The year the car was manufactured or first registered, ranging from 2000 to 2021	2000 – 2021
`Engine Size`	Numerical	Engine displacement in litres (L) — a higher value generally means a more powerful engine	1.0 L – 4.5 L
`Mileage`	Numerical	Total distance the car has been driven (km). Higher mileage usually reduces price	56 km – 200,000 km

Column	Data Type	Description	Sample Values
`Fuel Type`	Categorical	The type of fuel the car uses: Petrol, Diesel, Hybrid, or Electric	Petrol / Diesel / Electric / Hybrid
`Transmission`	Categorical	The gearbox system: Manual or Automatic	Manual / Automatic
`Price`	Numerical	**The selling price of the vehicle — THIS IS THE TARGET VARIABLE you must predict**	~£6,700 – £41,800

Data Distribution Summary

The table below gives you an overview of what values exist in each column and how they are spread. Read this carefully before starting your Exploratory Data Analysis — it will help you know what to look for.

Feature	Range / Count	Distribution Notes
Make	5 brands	Ford ~23%, Audi ~21%, BMW, Honda, Toyota
Model	5 variants	Model B & C most common (~21% each)
Year	2000 – 2021	Fairly evenly spread; 133 cars from year 2000
Engine Size	1.0 L – 4.5 L	Spread across full range, peak at 3.8–4.1 L
Mileage	56 – 200,000	Very evenly distributed — no clustering
Fuel Type	4 types	Diesel 34%, Petrol 33%, Electric + Hybrid 33%
Transmission	2 types	Manual 51%, Automatic 49% — nearly equal
Price □	~£6.7K – £41.8K	Most cars between £17K – £35K (bell-shaped)

Project Tasks

Complete all 6 sections below inside ONE Jupyter Notebook (.ipynb). Each section must be clearly labelled with a Markdown heading. Your notebook should tell a clear story — from loading the data, all the way to evaluating your model.

#	Section	What to Do
1	**Project Setup**	Load the CSV file, import all libraries, and display .head(), .info(), and .describe().
2	**Exploratory Data Analysis**	Visualise distributions, check for missing values, and produce a

		correlation analysis with at least 6 charts.
3	**Data Preprocessing**	Encode categorical columns, scale numerical features, and perform a train-test split (80/20, random_state=42).
4	**Build an ML Pipeline**	Create a scikit-learn Pipeline (Imputer → Encoder/Scaler → LinearRegression) and train it on the training set.
5	**Model Evaluation**	Report MAE, MSE, RMSE, and R^2. Run 5-fold cross-validation. Plot Actual vs Predicted values.
6	**Summary & Reflection**	Write a professional Markdown summary: key EDA findings, model performance in plain English, and 3 next-steps ideas.